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Composite electrode

Abstract

An apparatus is disclosed that includes an active storage layer including: a network of carbon nanotubes defining void spaces; and a carbonaceous material located in the void spaces and bound by the network of carbon nanotubes. In some cases, the active layer provides energy storage, e.g., in an ultracapacitor device.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) The present application is a continuation of U.S. patent application Ser. No. 16/681,293, filed Nov. 12, 2019, now U.S. Pat. No. 10,600,582, granted Mar. 24, 2020, which is a continuation of U.S. patent application Ser. No. 16/427,546 filed on May 31, 2019, which is a continuation-in-part of PCT/US2017/064152 filed on Dec. 1, 2017, which claims the benefit of U.S. Provisional Application No. 62/429,727, filed Dec. 2, 2016, the entire contents of which are incorporated herein by reference.

BACKGROUND

(1) Carbon nanotubes (hereinafter referred to also as “CNTs”) are carbon structures that exhibit a variety of properties. Many of the properties suggest opportunities for improvements in a variety of technology areas. These technology areas include electronic device materials, optical materials as well as conducting and other materials. For example, CNTs are proving to be useful for energy storage in capacitors.

(2) However, CNTs are typically expensive to produce and may present special challenges during electrode manufacturing. Accordingly, there is a need for an electrode material that exhibits the advantageous properties of CNTs while mitigating the amount of CNTs included in the material.

SUMMARY

(3) The applicants have developed a composite electrode structure that exhibits advantageous properties. In some embodiments, the electrode exhibits the advantageous properties of CNTs while mitigating the amount of CNTs included in the material, e.g., to less than 10% by weight.

(4) Electrodes of the type described herein may be used in ultracapacitors to provide high performance (e.g., high operating voltage, high operating temperature, high energy density, high power density, low equivalent series resistance, etc.).

(5) In one aspect, an apparatus is disclosed including an active storage layer including a network of carbon nanotubes defining void spaces; and a carbonaceous material located in the void spaces and bound by the network of carbon nanotubes, wherein the active layer is configured to provide energy storage.

(6) In some embodiments, the active layer is substantially free from binding agents. In some embodiments, the active layer consists of or consists essentially of carbonaceous material. In some embodiments, the active layer is bound together by electrostatic forces between the carbon nanotubes and the carbonaceous material. In some embodiments, the carbonaceous material includes activated carbon.

(7) In some embodiments, the carbonaceous material includes nanoform carbon other than carbon nanotubes.

(8) In some embodiments, the network of carbon nanotubes makes up less than 50% by weight of the active layer, less than 10% by weight of the active layer, less than 5% by weight of the active layer, or less than 1% by weight of the active layer.

(9) Some embodiments include an adhesion layer, e.g., a layer consisting of or consisting essentially of carbon nanotubes. In some embodiments the adhesion layer is disposed between the active layer and an electrically conductive layer.

(10) In some embodiments, a surface of the conductive layer facing the adhesion layer includes a roughened or textured portion. In some embodiments, a surface of the conductive layer facing the adhesion layer includes a nanostructured portion. In some embodiments, the nanostructured portion includes carbide “nanowhiskers”. These nanowhiskers are thin elongated structures (e.g., nanorods) that extend generally away from the surface of the conductor layer **102**. The nanowhiskers may have a radial thickness of less than 100 nm, 50 nm, 25, nm, 10 nm, or less, e.g., in the range of 1 nm to 100 nm or any subrange thereof. The nanowhisker may have a longitudinal length that is

several to many times its radial thickness, e.g., greater than 20 nm, 50 nm, 100 nm, 200 nm, 300, nm, 400, nm, 500 nm, 1 μ m, 5 μ m, 10 μ m, or more, e.g., in the range of 20 nm to 100 μ m or any subrange thereof.

(11) In some embodiments, the active layer has been annealed to reduce the presence of impurities.

(12) In some embodiments, active layer has been compressed to deform at least a portion of the network of carbon nanotubes and carbonaceous material.

(13) Some embodiments include an electrode including the active layer. Some embodiments include an ultracapacitor including the electrode, In some embodiments, the ultracapacitor has an operating voltage greater than 1.0 V, 2.0 V, 2.5 V 3.0 V, 3.1 V, 3.2 V, 3.5 V, 4.0 V or more.

(14) In some embodiments, the ultracapacitor has a maximum operating temperature of at least 250 C at an operating voltage of at least 1.0 V for a lifetime of at least 1,000 hours. In some embodiments, the ultracapacitor has a maximum operating temperature of at least 250 C at an operating voltage of at least 2.0 V for a lifetime of at least 1,000 hours. In some embodiments, the ultracapacitor has a maximum operating temperature of at least 250 C at an operating voltage of at least 3.0 V for a lifetime of at least 1,000 hours. In some embodiments, the ultracapacitor has a maximum operating temperature of at least 250 C at an operating voltage of at least 4.0 V for a lifetime of at least 1,000 hours. In some embodiments, the ultracapacitor has a maximum operating temperature of at least 300 C at an operating voltage of at least 1.0 V for a lifetime of at least 1,000 hours. In some embodiments, the ultracapacitor has a maximum operating temperature of at least 300 C at an operating voltage of at least 2.0 V for a lifetime of at least 1,000 hours. In some embodiments, the ultracapacitor has a maximum operating temperature of at least 300 C at an operating voltage of at least 3.0 V for a lifetime of at least 1,000 hours. In some embodiments, the ultracapacitor has a maximum operating temperature of at least 300 C at an operating voltage of at least 4.0 V for a lifetime of at least 1,000 hours.

(15) In another aspect, a method including: dispersing carbon nanotubes in a solvent to form a dispersion; mixing the dispersion with carbonaceous material to form a slurry; applying the slurring in a layer; and drying the slurry to substantially remove the solvent to form an active layer including a network of carbon nanotubes defining void spaces and a carbonaceous material located in the void spaces and bound by the network of carbon nanotubes. Some embodiment include forming or applying a layer of carbon nanotubes to provide an adhesion layer on a conductive layer.

(16) In some embodiments, the applying step including applying the slurry onto the adhesion layer.

(17) Various embodiments may include any of the forgoing elements or features, or any elements or features described herein either alone or in any suitable combination.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic of an electrode.

(2) FIG. 2 is an illustration of a detailed view of an active layer of an electrode.

(3) FIG. 3 is a schematic of a two-sided electrode.

(4) FIG. 4 is a flow chart illustrating a method of making an active layer for an electrode.

(5) FIG. 5 is a flow chart illustrating a method of making an adhesion layer for an electrode.

(6) FIG. 6 is a schematic diagram of an exemplary mixing apparatus.

(7) FIG. 7A is a schematic diagram of coating apparatus featuring a slot die.

(8) FIG. 7B is a schematic diagram of coating apparatus featuring a doctor blade.

(9) FIG. 8A is a schematic of an ultracapacitor.

(10) FIG. 8B is a schematic of an ultracapacitor without a separator.

DETAILED DESCRIPTION

(11) Referring to FIG. 1, an exemplary embodiment of an electrode **100** is disclosed for use in an energy storage device, such as an ultracapacitor or battery. The electrode includes an electrically conductive layer **102** (also referred to herein as a current collector), an adhesion layer **104**, and an active layer **106**. When used in an ultracapacitor of the type described herein, the active layer **106** may act as energy storage media, for example, by providing a surface interface with an electrolyte (not shown) for formation of an electric double layer (sometimes referred to in the art as a Helmholtz layer). In some embodiments, the adhesion layer **104** may be omitted, e.g., in cases where the active layer **106** exhibits good adhesion to the electrically conductive layer **102**.

(12) In some embodiments, the active layer **106** may be thicker than the adhesion layer **104**, e.g., 1.5, 2.0, 5.0, 10, 20, 30, 40, 50, 60, 70, 80, 90, 100, 500, 1,000 or more times the thickness of the adhesion layer **104**. For example, in some embodiments, the thickness of the active layer **106** may be in the range of 1.5 to 1,000 times the thickness of the adhesion layer **104** (or any subrange thereof, such as 5 to 100 times). For example, in some embodiments the active layer **106** may have a thickness in the range of 0.5 to 2500 μm or any subrange thereof, e.g., 5 μm to 150 μm . In some embodiments the adhesion layer **104** may have a thickness in the range of 0.5 μm to 50 μm or any subrange thereof, e.g., 1 μm to 5 μm .

(13) Referring to FIG. 2, in some embodiments, the active layer **106** is comprised of carbonaceous material **108** (e.g., activated carbon) bound together by a matrix **110** of CNTs **112** (e.g., a webbing or network formed of CNTs). In some embodiments, e.g., where the length of the CNTs is longer than the thickness of the active layer **106**, the CNTs **112** forming the matrix **110** may lie primarily parallel to a major surface of the active layer **106**. Not that although as shown the CNTs **112** form straight segments, in some embodiments, e.g., where longer CNTs are used, the some or all of the CNTs may instead have a curved or serpentine shape. For example, in cases where the carbonaceous material **108** includes lumps of activated carbon, the CNTs **112** may curve and wind between the lumps.

(14) In some embodiments, the active layer is substantially free of any other binder material, such as polymer materials, adhesives, or the like. In other words, in such embodiments, the active layer is substantially free from any material other than carbon. For example, in some embodiments, the active layer may be at least about 90 wt %, 95 wt %, 96 wt %, 97 wt %, 98 wt %, 99 wt %, 99.5 wt %, 99.9 wt %, 99.99 wt %, 99.999 wt %, or more elemental carbon by mass. Despite this, the matrix **110** operated to bind together the carbonaceous material **108**, e.g., to maintain the structural integrity of the active layer **106** without flaking, delamination, disintegration, or the like.

(15) It has been found that use of an active layer substantially free of any non-carbon impurities substantially increases the performance of the active layer in the presence of high voltage differentials, high temperatures, or both. Not wishing to be bound by theory, it is believed that the lack of impurities prevents the occurrence of unwanted chemical side reactions which otherwise would be promoted in high temperature or high voltage conditions.

(16) As noted above, in some embodiments, the matrix **110** of carbon nanotubes provides a structural framework for the active layer **106**, with the carbonaceous material **108** filling the spaces between the CNTs **112** of the matrix **110**. In some embodiments, electrostatic forces (e.g., Van Der Waals forces) between the CNTs **112** within the matrix **110** and between the matrix **112** and the other carbonaceous material **108** may provide substantially all of the binding forces maintaining the structural integrity of the layer.

(17) In some embodiments, the CNTs **112** may include single wall nanotubes (SWNT), double wall nanotubes (DWNT), or multiwall nanotubes (MWNT), or mixtures thereof. Although a matrix **110** of individual CNTs **112** is shown, in some embodiments, the matrix may include interconnected bundles, clusters or aggregates of CNTs. For example, in some embodiments where the CNTs are initially formed as vertically aligned, the matrix may be made up at least in part of brush like bundles of aligned CNTs.

(18) In order to provide some context for the teachings herein, reference is first made to U.S. Pat.

No. 7,897,209, entitled “Apparatus and Method for Producing Aligned Carbon Nanotube Aggregate.” The foregoing patent (the “209 patent”) teaches a process for producing aligned carbon nanotube aggregate. Accordingly, the teachings of the '209 patent, which are but one example of techniques for producing CNTs in the form of an aligned carbon nanotube aggregate, may be used to harvest CNTs referred to herein. Advantageously, the teachings of the '209 patent may be used to obtain long CNTs having high purity. In other embodiments, any other suitable method known in the art for producing CNTs may be used.

(19) In some embodiments the active layer **106** may be formed as follows. A first solution (also referred to herein as a slurry) is provided that includes a solvent and a dispersion of carbon nanotubes, e.g., vertically aligned carbon nanotubes. A second solution (also referred to herein as a slurry) may be provided that includes a solvent with carbon disposed therein. This carbon addition includes at least one form of material that is substantially composed of carbon. Exemplary forms of the carbon addition include, for example, at least one of activated carbon, carbon powder, carbon fibers, rayon, graphene, aerogel, nanohorns, carbon nanotubes and the like. While in some embodiments, the carbon addition is formed substantially of carbon, it is recognized that in alternative embodiments the carbon addition may include at least some impurities, e.g., additives included by design.

(20) In some embodiments, forming the first and/or second solution include introducing mechanical energy into the mixture of solvent and carbon material, e.g., using a sonicator (sometimes referred to as a sonifier) or other suitable mixing device (e.g., a high shear mixer). In some embodiments, the mechanical energy introduced into the mixture per kilogram of mixture is at least 0.4 kWh/kg, 0.5 kWh/kg, 0.6 kWh/kg, 0.7 kWh/kg, 0.8 kWh/kg, 0.9 kWh/kg, 1.0 kWh/kg, or more. For example, the mechanical energy introduced into the mixture per kilogram of mixture may be in the range of 0.4 kWh/kg to 1.0 kWh/kg or any subrange thereof such as 0.4 kWh/kg to 0.6 kWh/kg.

(21) In some embodiments, the solvents used may include an anhydrous solvent. For example, the solvent may include at least one of ethanol, methanol, isopropyl alcohol, dimethyl sulfoxide, dimethylformamide, acetone, acetonitrile, and the like.

(22) As noted above, the two solutions may be subjected to “sonication” (physical effects realized in an ultrasonic field). With regard to the first solution, the sonication is generally conducted for a period that is adequate to tease out, fluff or otherwise parse the carbon nanotubes. With regard to the second solution, the sonication is generally conducted for a period that is adequate to ensure good dispersion or mixing of the carbon additions within the solvent. In some embodiments, other techniques for imparting mechanical energy to the mixtures may be used in addition or alternative to sonication, e.g., physical mixing using stirring or impeller.

(23) Once one or both of the first solution and the second solution have been adequately sonicated, they are then mixed together, to provide a combined solution and may again be sonicated. Generally, the combined mixture is sonicated for a period that is adequate to ensure good mixing of the carbon nanotubes with the carbon addition. This second mixing (followed by suitable application and drying steps as described below) results in the formation of the active layer **106** containing the matrix **110** of CNTs **112**, with the carbon addition providing the other carbonaceous material **108** filling the void spaces of the matrix **110**.

(24) In some embodiments, mechanical energy may be introduced to the combined mixture using a sonicator (sometimes referred to as a sonifier) or other suitable mixing device (e.g., a high shear mixer). In some embodiments, the mechanical energy into the mixture per kilogram of mixture is at least 0.4 kWh/kg, 0.5 kWh/kg, 0.6 kWh/kg, 0.7 kWh/kg, 0.8 kWh/kg, 0.9 kWh/kg, 1.0 kWh/kg, or more. For example, the mechanical energy introduced into the mixture per kilogram of mixture may be in the range of 0.4 kWh/kg to 1.0 kWh/kg or any subrange thereof such as 0.4 kWh/kg to 0.6 kWh/kg.

(25) In some embodiments, the combined slurry may be cast wet directly onto the adhesion layer

104 or the conductive layer **102**, and dried (e.g., by applying heat or vacuum or both) until substantially all of the solvent and any other liquids have been removed, thereby forming the active layer **106**. In some such embodiments it may be desirable to protect various parts of the underlying layers (e.g., an underside of a conductive layer **102** where the current collector is intended for two sided operation) from the solvent, e.g., by masking certain areas, or providing a drain to direct the solvent.

(26) In other embodiments, the combined slurry may be dried elsewhere and then transferred onto the adhesion layer **104** or the conductive layer **102** to form the active layer **106**, using any suitable technique (e.g., roll-to-roll layer application). In some embodiments the wet combined slurry may be placed onto an appropriate surface and dried to form the active layer **106**. While any material deemed appropriate may be used for the surface, exemplary material includes PTFE as subsequent removal from the surface is facilitated by the properties thereof. In some embodiments, the active layer **106** is formed in a press to provide a layer that exhibits a desired thickness, area and density.

(27) In some embodiments, the average length of the CNTs **112** forming the matrix **110** may be at least 0.1 μm , 0.5 μm , 1 μm , 5 μm , 10 μm , 50 μm , 100 μm , 200 μm , 300 μm , 400 μm , 500 μm , 600 μm , 700 μm , 800 μm , 900 μm , 1,000 μm or more. For example, in some embodiments, the average length of the CNTs **112** forming the matrix **110** may be in the range of 1 μm to 1,000 μm , or any subrange thereof, such as 1 μm to 600 μm . In some embodiments, more than 50%, 60%, 70%, 80%, 90%, 95%, 99% or more of the CNTs **112** may have a length within 10% of the average length of the CNTs **112** making up the matrix **110**.

(28) In various embodiments, the other carbonaceous material **108** can include carbon in a variety forms, including activated carbon, carbon black, graphite, and others. The carbonaceous material can include carbon particles, including nanoparticles, such as nanotubes, nanorods, graphene in sheet, flake, or curved flake form, and/or formed into cones, rods, spheres (buckyballs) and the like.

(29) Applicants have found unexpected result that an active layer of the type herein can provide exemplary performance (e.g., high conductivity, low resistance, high voltage performance, and high energy and power density) even when the mass fraction of CNTs in the layer is quite low. For example, in some embodiments, the active layer may be at least about 50 wt %, 60 wt %, 70 wt %, 75 wt %, 80 wt %, 85 wt %, 90 wt %, 95 wt %, 96 wt %, 97 wt %, 98 wt %, 99 wt %, 99.5 wt %, or more elemental carbon in a form other than CNT (e.g., activated carbon). In particular, for certain applications involving high performance ultracapacitors, active layers **106** that are in the range of 95 wt % to 99 wt % activated carbon (with the balance CNTs **112**), have been shown to exhibit excellent performance.

(30) In some embodiments, the matrix **110** of CNTs **112** form an interconnected network of highly electrically conductive paths for current flow (e.g. ion transport) through the active layer **106**. For example, in some embodiments, highly conductive junctions may occur at points where CNTs **112** of the matrix **110** intersect with each other, or where they are in close enough proximity to allow for quantum tunneling of charge carriers (e.g., ions) from one CNT to the next. While the CNTs **112** may make up a relatively low mass fraction of the active layer (e.g., less than 10 wt %, 5 wt %, 4 wt %, 3 wt %, 2 wt %, 1 wt % or less, e.g., in the range of 0.5 wt % to 10 wt % or any subrange thereof such as 1 wt % to 5.0 wt %), the interconnected network of highly electrically conductive paths formed in the matrix **110** may provide long conductive paths to facilitate current flow within and through the active layer **106** (e.g. conductive paths on the order of the thickness of the active layer **106**).

(31) For example, in some embodiments, the matrix **110** may include one or more structures of interconnected CNTs, where the structure has an overall length in along one or more dimensions longer than 2, 3, 4, 5, 10, 20, 50, 100, 500, 1,000, 10,000 or more times the average length of the component CNTs making up the structure. For example, in some embodiments, the matrix **110** may include one or more structures of interconnected CNTs, where the structure has an overall in the

range of 2 to 10,000 (or any subrange thereof) times the average length of the component CNTs making up the structure. For example, in some embodiments the matrix **110** may include highly conductive pathways having a length greater than 100 μm , 500 μm , 1,000 μm , 10,000 μm or more, e.g., in the range of 100 μm -10,000 μm of any subrange thereof.

(32) As used herein, the term “highly conductive pathway” is to be understood as a pathway formed by interconnected CNTs having an electrical conductivity higher than the electrical conductivity of the other carbonaceous material **108** (e.g., activated carbon), surrounding that matrix **110** of CNTs **112**.

(33) Not wishing to be bound by theory, in some embodiments the matrix **110** can be characterized as an electrically interconnected network of CNT exhibiting connectivity above a percolation threshold. Percolation threshold is a mathematical concept related to percolation theory, which is the formation of long-range connectivity in random systems. Below the threshold a so called “giant” connected component of the order of system size does not exist; while above it, there exists a giant component of the order of system size.

(34) In some embodiments, the percolation threshold can be determined by increasing the mass fraction of CNTs **112** in the active layer **106** while measuring the conductivity of the layer, holding all other properties of the layer constant. In some such cases, the threshold can be identified with the mass fraction at which the conductivity of the layer sharply increases and/or the mass fraction above which the conductivity of the layer increases only slowly with increases with the addition of more CNTs. Such behavior is indicative of crossing the threshold required for the formation of interconnected CNT structures that provide conductive pathways with a length on the order of the size of the active layer **106**.

(35) Returning to FIG. **1**, in some embodiments, one or both of the active layer **106** and the adhesion layer **104** may be treated by applying heat to remove impurities (e.g., functional groups of the CNTs, and impurities such as moisture, oxides, halides, or the like). For example, in some embodiments, one or both of the layers can be heated to at least 100 C, 150 C, 200 C, 250 C, 300 C, 350 C, 400 C, 450 C, 500 C or more for at least 1 minute, 5 minutes, 10 minutes, 30 minutes, 1 hour, 2 hours, 3 hours, 12 hours, 24 hours, or more. For example, in some embodiments the layers may be treated to reduce moisture in the layer to less than 1,000 ppm, 500 ppm, 100 ppm, 10 ppm, 1 ppm, 0.1 ppm or less.

(36) Returning to FIG. **1**, in some embodiments, the adhesion layer **104** may be formed of carbon nanotubes. For example, in some embodiments, the adhesion layer **104** may be at least about 50%, 75%, 80%, 90%, 95%, 96%, 97%, 98%, 99%, 99.5%, 99.9%, 99.99%, 99.999% by mass CNTs. In some embodiments, the CNTs may be grown directly on the conductive layer **102**, e.g., using the chemical vapor deposition techniques such as those described in U.S. Patent Pub. No. 2013/0045157 entitled “In-line Manufacture of Carbon Nanotubes” and published Feb. 21, 2013. In some embodiments, the CNTs may be transferred onto the conductive layer **102**, e.g., using wet or dry transfer processes, e.g., of the type described e.g., in U.S. Patent Pub. No. 2013/0044405 entitled “High Power and High Energy Electrodes Using Carbon Nanotubes” and published Feb. 21, 2013. In some embodiments, the adhesion layer **104** adheres to the overlying active layer **106** using substantially only electrostatic forces (e.g., Van Der Waals attractions) between the CNTs of the adhesion layer **104** and the carbon material and CNTs of the active layer **106**.

(37) In some embodiments, the CNTs of the adhesion layer **104** may include single wall nanotubes (SWNT), double wall nanotubes (DWNT), or multiwall nanotubes (MWNT), or mixtures thereof. In some embodiments the CNTs may be vertically aligned. In one particular embodiment, the CNTs of the adhesion layer **104** may be primarily or entirely SWNTs and/or DWNTs, while the CNTs of the active layer **106** are primarily or entirely MWNTs. For example, in some embodiments, the CNTs of the adhesion layer **104** may be at least 75%, at least 90%, at least 95%, at least 99% or more SWNT or at least 75%, at least 90%, at least 95%, at least 99% or more DWNT. In some embodiments, the CNTs of the active layer **106** may be at least 75%, at least 90%, at least

95%, at least 99% or more MWNT.

(38) In some embodiments, the adhesion layer **104** may be formed by applying pressure to a layer of carbonaceous material. In some embodiments, this compression process alters the structure of the adhesion layer **104** in a way that promotes adhesion to the active layer **106**. For example, in some embodiments pressure may be applied to layer comprising a vertically aligned array of CNT or aggregates of vertically aligned CNT, thereby deforming or breaking the CNTs.

(39) In some embodiments, the adhesion layer may be formed by casting a wet slurry of CNTs (with or without additional carbons) mixed with a solvent onto the conductive layer **102**. In various embodiments, similar techniques to those described above for the formation of the active layer **106** from a wet slurry may be used.

(40) In some embodiments, mechanical energy may be introduced to the wet slurry using a sonicator (sometimes referred to as a sonifier) or other suitable mixing device (e.g., a high shear mixer). In some embodiments, the mechanical energy into the mixture per kilogram of mixture is at least 0.4 kWh/kg, 0.5 kWh/kg, 0.6 kWh/kg, 0.7 kWh/kg, 0.8 kWh/kg, 0.9 kWh/kg, 1.0 kWh/kg, or more. For example, the mechanical energy introduced into the mixture per kilogram of mixture may be in the range of 0.4 kWh/kg to 1.0 kWh/kg or any subrange thereof such as 0.4 kWh/kg to 0.6 kWh/kg.

(41) In some embodiments, the solid carbon fraction of the wet slurry may be less than 10 wt %, 5 wt %, 4 wt %, 3 wt %, 2 wt %, 1 wt %, 0.5 wt %, 0.1 wt % or less, e.g., in the range of 0.1 wt % to 10 wt % or any subrange thereof such as 0.1 wt % to 2 wt %.

(42) In various embodiments, the conductive layer **102** may be made of a suitable electrically conductive material such as a metal foil (e.g., an aluminum foil). In some embodiments, the surface of the conductive layer **102** may be roughened, patterned, or otherwise texturized, e.g., to promote adhesion to the adhesion layer **104** and good electrical conductance from the active layer **106**. For example, in some embodiments, the conductive layer may be etched (e.g., mechanically or chemically). In some embodiments, the conductive layer **102** may have a thickness in the range of 1 μm to 1,000 μm or any subrange thereof such as 5 μm to 50 μm .

(43) In some embodiments, the conductive layer **102** may include a nanostructured surface. For example, as described in International Pub. No. WO 2016/057983 entitled “Nanostructured Electrode for Energy Storage Device” published Apr. 14, 2016, the conductive layer may have a top surface that includes nanoscale features such as whiskers (e.g., carbide whiskers) that promote adhesion to the adhesion layer **104** and good electrical conductance from the active layer **106**. An exemplary current collector is the current collector available from Toyo Aluminum K.K. under the trade name TOYAL-CARBO®.

(44) In some embodiments, one or both of the active layer **106** and the adhesion layer **104** may be treated by applying heat and/or vacuum to remove impurities (e.g., functional groups of the CNTs, and impurities such as moisture, oxides, halides, or the like).

(45) In some embodiments, one or both of the active the active layer **106** and the adhesion layer **104** may be compressed, e.g., to break some of the constituent CNTs or other carbonaceous material to increase the surface area of the respective layer. In some embodiments, this compression treatment may increase one or more of adhesion between the layers, ion transport rate within the layers, and the surface area of the layers. In various embodiments, compression can be applied before or after the respective layer is applied to or formed on the electrode **100**.

(46) In some embodiments, the adhesion layer **104** may be omitted, such that the active layer **106** is disposed directly on the conductive layer **102**.

(47) Referring to FIG. 3, in some embodiments, the electrode **100** may be double sided, with an adhesion layer **104** and active layer **106** formed on each of two opposing major surfaces of the conductive layer **102**. In some embodiments, the adhesion layer **104** may be omitted on one or both sides of the two-sided electrode **100**.

(48) Referring to FIG. 4, an exemplary embodiment of method **200** of making the active layer **106**

of electrode **100** is described. In step **201**, CNTs are dispersed in a solvent to form a dispersion of CNTs. In some embodiments, the dispersion may be formed using any of the techniques described in U.S. Patent Pub. No. 2013/0044405 entitled “High Power and High Energy Electrodes Using Carbon Nanotubes” published Feb. 21, 2013 including stirring, sonication, or a combination of the two. In various embodiments, any suitable solvent may be used, including, for example, ethanol, methanol, isopropyl alcohol, dimethyl sulfoxide, dimethylformamide, acetone, acetonitrile, and the like. In general, it is advantageous to choose a solvent that will be substantially eliminated in the drying step **204** described below, e.g., using heat and/or vacuum drying techniques.

(49) In some embodiments, the mixture of CNTs and solvents may be passed through a filter, e.g., an array of micro channels (e.g., having channels with diameters on the order of the radial size of the CNTs) to help physically separate the CNTs and promote dispersion.

(50) In some embodiments, the CNT dispersion may be formed without the addition of surfactants, e.g., to avoid the presence of impurities derived from these surfactants at the completion of the method **200**.

(51) In step **202**, the CNT dispersion is mixed with carbonaceous material (e.g., activated carbon) to form a slurry. In some embodiments, the slurry may be formed using any of the techniques described in U.S. Patent Pub. No. 2013/0044405, including stirring, sonication, or a combination of the two. In some embodiments, the slurry may have solid carbon fraction of less than 20 wt %, 15 wt %, 10 wt %, 5 wt %, 2 wt %, 1 wt %, or less, e.g., in the range of 1 wt % to 20 wt % or any subrange thereof such as 4% to 6%. The mass ratio of CNTs to other carbonaceous material in the slurry may be less than 1:5, 1:10, 1:15, 1:20, 1:50, 1:100, or less, e.g., in the range of 1:10 to 1:20 or any subrange thereof.

(52) In step **203**, the slurry is applied to either the adhesion layer **104** or, if the adhesion layer **104** is omitted, the conductive layer **102** of the electrode **100**. In some embodiments, the slurry may be formed into a sheet, and coated onto the electrode. For example, in some embodiments, slurry may be applied to through a slot die to control the thickness of the applied layer. In other embodiments, the slurry may be applied to the conductive layer **102**, and then leveled to a desired thickness, e.g., using a doctor blade.

(53) In some embodiments, the slurry may be compressed (e.g., using a calendaring apparatus) before or after being applied to the electrode **100**. In some embodiments, the slurry may be partially or completely dried (e.g., by applying heat, vacuum or a combination thereof) during this step **203**.

(54) In step **204**, if the slurry has not dried, or has been only partially dried during step **203**, the slurry applied to the electrode is fully dried, (e.g., by applying heat, vacuum or a combination thereof). In some embodiments, substantially all of the solvent (and any other non-carbonaceous material such as dispersing agents) is removed from the active layer **106**. In some embodiments, if impurities remain following the drying step, and additional step of heating (e.g. baking or annealing) the layer may be performed. For example, in some embodiments, one or both of the active the active layer **106** and the adhesion layer **104** may be treated by applying heat to remove impurities (e.g., functional groups of the CNTs, and impurities such as moisture, oxides, halides, or the like).

(55) Referring to FIG. 5, an exemplary embodiment of method **300** of making the adhesion layer **104** of electrode **100** is described. In step **301**, CNTs are dispersed in a solvent to form a dispersion of CNTs. In some embodiments, the dispersion may be formed using any of the techniques described in U.S. Patent Pub. No. 2013/0044405 including stirring, sonication, or a combination of the two. In various embodiments, any suitable solvent may be used, including, for example an organic solvent such as isopropyl alcohol, acetonitrile or propylene carbonate. In general, it is advantageous to choose a solvent that will be substantially eliminated in the drying step **304** described below.

(56) In some embodiments, the mixture of CNTs and solvents may be passed through a filter, e.g., an array of micro channels (e.g., having channels with diameters on the order of the radial size of

the CNTs) to help physically separate the CNTs and promote dispersion.

(57) In some embodiments, the CNT dispersion may be formed without the addition of surfactants, e.g., to avoid the presence of impurities derived from these surfactants at the completion of the method **300**.

(58) In step **302**, the CNT dispersion may optionally be mixed with additional carbonaceous material (e.g., activated carbon) to form a slurry. In some embodiments, the additional carbonaceous material may be omitted, such that the slurry is made up of CNTs dispersed in a solvent. In some embodiments, the slurry may have solid fraction of less than 5 wt %, 4 wt %, 3 wt %, 2 wt %, 1 wt %, 0.5 wt %, 0.1 wt %, or less, e.g., in the range of 0.1 to 5 wt % or any subrange thereof.

(59) In step **303**, the slurry is applied to the conductive layer **102** of the electrode **100**. In some embodiments, the slurry may be coated onto the electrode. For example, in some embodiments, slurry may be applied to through a slot die to control the thickness of the applied layer. In other embodiments, the slurry may be applied to the conductive layer **102**, and then leveled to a desired thickness, e.g., using a doctor blade.

(60) In some embodiments, the slurry may be compressed (e.g., using a calendaring apparatus) before or after being applied to the electrode **100**. In some embodiments, the slurry may be partially or completely dried (e.g., by applying heat, vacuum or a combination thereof) during this step **303**.

(61) In step **304**, if the slurry has not dried, or has been only partially dried during step **203**, the slurry applied to the electrode is fully dried, (e.g., by applying heat, vacuum or a combination thereof). In some embodiments, substantially all of the solvent (and any other non-carbonaceous material such as dispersing agents) is removed from the active layer **106**. In some embodiments, if impurities remain following the drying step, and additional step of heating (e.g. baking or annealing) the layer may be performed. For example, in some embodiments, one or both of the active the active layer **106** and the adhesion layer **104** may be treated by applying heat to remove impurities (e.g., functional groups of the CNTs, and impurities such as moisture, oxides, halides, or the like).

(62) In some embodiments, the method **300** for forming an adhesion layer **104** and method **200** for forming an active layer **106** may be performed in series to successively form the adhesion layer **104** followed by the overlaying active layer **106**. In some embodiments, the foregoing methods may be repeated, e.g., to form a two-sided electrode of the type described herein.

(63) Advantageously, in some embodiments, the method **300** for forming an adhesion layer **104** and/or method **200** for forming an active layer **106** may be implemented as a roll-to-roll processes, e.g., to allow volume production of electrode sheets several tens of meters long or more.

(64) FIG. **6** shows an exemplary mixing apparatus **400** for implementing the method **300** for forming an adhesion layer **104** and/or method **200** for forming an active layer **106**. In the interest of brevity, the apparatus **400** will be described for use in forming active layer **106** using method **200**. However, as will be apparent to one skilled in the art, the apparatus **400** can easily be configured to implement the method **300** for forming an adhesion layer **104**.

(65) The apparatus **400** includes a mixing vessel **401**. The mixing vessel receives a slurry composed of a solvent, carbon nanotubes, and (optionally) additional carbonaceous material of the type described above. In some embodiments, this slurry (or components thereof) may be initially formed in the mixing vessel **401**. In other embodiments, the slurry may be formed elsewhere and then transferred to the mixing vessel **401**.

(66) In some embodiments the mixing vessel **401** may include one or more mechanisms for mixing the slurry, such as an impeller or high sheer mixer. In some embodiments, a mixing mechanism may be provided which is capable of stirring the slurry at a controlled rate, e.g., of up to 1000 rotations per minute (RPM) or more. In some embodiments, the mixing vessel may include one or more devices for applying mechanical energy to the slurry, such as a sonicator, mixer (e.g., a high shear mixer), homogenizer, or any other suitable device known in the art. In some embodiments,

the mixing vessel may be temperature controlled, e.g., using one or more heating and/or cooling elements such as electric heaters, tubing for circulating chilled water, or any other such devices known in the art.

(67) Slurry from the mixing vessel **401** may be circulated through a flow line **402**, e.g. a pipe or tubing, using a pump **403**. Pump **403** may be any suitable configuration, such as a peristaltic pump. A flow meter **404** may be provided to measure the rate of slurry flow through the flow line **402**. A filter **405** may be provided to filter the slurry flowing through the flow line **402**, e.g., to remove clumps of solid material having a size above a desired threshold.

(68) In some embodiments, e.g., where mixing vessel **401** does not include a sonicator, an in-line sonicator **406** may be provided to sonicate slurry flowing through the flow line **402**. For example, in some embodiments a flow through sonicator such as the Branson Digital SFX-450 sonicator available commercially from Thomas Scientific of 1654 High Hill Road Swedesboro, N.J. 08085 U.S.A may be used.

(69) In some embodiments, a temperature control device **407**, such as a heat exchanger arranged in a sleeve disposed about the flow line **402**, is provided to control the temperature of the slurry flowing through the flow line **402**.

(70) In some embodiments a valve **408** is provided which can be selectively controlled to direct a first portion of the slurry flowing through flow line **402** to be recirculated back to the mixing vessel **401**, while a second portion is output externally, e.g., to a coating apparatus **500**.

(71) In some embodiments, a sensor **409** such as a pressure sensor or flow rate sensor is provided to sense one or more aspects of the output portion of slurry.

(72) In various embodiments any or all of the elements of apparatus **400** may be operatively connected to one or more computing devices to provide for automatic monitoring and/or control of the mixing apparatus **400**. For example, the sonicator **406** may include digital controls for controlling its operating parameters such as power and duty cycle.

(73) In various embodiments, the coating apparatus **500** may be any suitable type known in the art. For example, FIG. 7A shows an exemplary embodiment of coating apparatus **500** featuring a slot die **501** that distributes slurry received from a source such as the mixing apparatus **400** through a distribution channel **502** onto a substrate **503** (e.g., the conductive layer **102**, either bare or already coated with adhesion layer **104**) which moves across a roller **504**. Setting the height of the slot die above the substrate **503** on the roller **504** and controlling the flow rate and/or pressure of the slurry in the channel **502** allows for control of the thickness and density of the applied coating. In some embodiments, channel **502** may include one or more reservoirs to help ensure consistent flow of slurry to provide uniform coating during operation.

(74) FIG. 7B shows an exemplary embodiment of coating apparatus **500** featuring a doctor blade **601** that levels slurry received from a source such as the mixing apparatus **400** that is applied through one or more applicators **602** (one is shown) onto a substrate **603** (e.g., the conductive layer **102**, either bare or already coated with adhesion layer **104**) which moves across a roller **604**. The direction of travel of the substrate **603** is indicated by the heavy dark arrow. Setting the height of the doctor blade **601** above the substrate **603** on the roller **604** and controlling the flow rate and/or pressure of the slurry through the applicator **602** allows for control of the thickness and density of the applied coating. Although a single doctor blade **601** is shown, multiple blades may be used, e.g., a first blade to set a rough thickness of the coating, and a second blade positioned down line from the first blade to provide fine smoothing of the coating.

(75) Further, disclosed herein are capacitors incorporating the electrode that provide users with improved performance in a wide range of temperatures. Such ultracapacitors may comprise an energy storage cell and an electrolyte system within an hermetically sealed housing, the cell electrically coupled to a positive contact and a negative contact, wherein the ultracapacitor is configured to operate at a temperatures within a temperature range between about -100 degrees Celsius to about 300 degrees Celsius or more, or any subrange thereof, e.g., -40 C to 200 C, -40 C

to 250 C, -40 C to 300 C, 0 C to 200 C, 0 C to 250 C, 0 C to 300 C. In some embodiments such ultracapacitors can operate at voltages of 1.0 V, 2.0 V, 3.0 V, 3.2 V, 3.5 V, 4.0 V, or more, e.g., for lifetimes exceeding 1,000 hours.

(76) As shown in FIGS. **8A** and **8B**, exemplary embodiments of a capacitor are shown. In each case, the capacitor is an “ultracapacitor **10**.” The difference between FIG. **8A** and FIG. **8B** is the inclusion of a separator in exemplary ultracapacitor **10** of FIG. **8A**. The concepts disclosed herein generally apply equally to any exemplary ultracapacitor **10**. Certain electrolytes of certain embodiments are uniquely suited to constructing an exemplary ultracapacitor **10** without a separator. Unless otherwise noted, the discussion herein applies equally to any ultracapacitor **10**, with or without a separator.

(77) The exemplary ultracapacitor **10** is an electric double-layer capacitor (EDLC). The EDLC includes at least one pair of electrodes **3** (where the electrodes **3** may be referred to as a negative electrode **3** and a positive electrode **3**, merely for purposes of referencing herein). When assembled into the ultracapacitor **10**, each of the electrodes **3** (which may each be an electrode **100** of the type shown in FIG. **1** above) presents a double layer of charge at an electrolyte interface. In some embodiments, a plurality of electrodes **3** is included (for example, in some embodiments, at least two pairs of electrodes **3** are included). However, for purposes of discussion, only one pair of electrodes **3** are shown. As a matter of convention herein, at least one of the electrodes **3** uses a carbon-based energy storage media **1** (e.g., the active layer **106** of electrode **100** shown in FIG. **1**), and it is assumed that each of the electrodes includes the carbon-based energy storage media **1**. It should be noted that an electrolytic capacitor differs from an ultracapacitor because metallic electrodes differ greatly (at least an order of magnitude) in surface area.

(78) Each of the electrodes **3** includes a respective current collector **2** (also referred to as a “charge collector”), which may be the conductive layer **102** of electrode **100** shown in FIG. **1**. In some embodiments, the electrodes **3** are separated by a separator **5**. In general, the separator **5** is a thin structural material (usually a sheet) used to separate the negative electrode **3** from the positive electrode **3**. The separator **5** may also serve to separate pairs of the electrodes **3**. Once assembled, the electrodes **3** and the separator **5** provide a storage cell **12**. Note that, in some embodiments, the carbon-based energy storage media **1** may not be included on one or both of the electrodes **3**. That is, in some embodiments, a respective electrode **3** might consist of only the current collector **2**. The material used to provide the current collector **2** could be roughened, anodized or the like to increase a surface area thereof. In these embodiments, the current collector **2** alone may serve as the electrode **3**. With this in mind, however, as used herein, the term “electrode **3**” generally refers to a combination of the energy storage media **1** and the current collector **2** (but this is not limiting, for at least the foregoing reason).

(79) At least one form of electrolyte **6** is included in the ultracapacitor **10**. The electrolyte **6** fills void spaces in and between the electrodes **3** and the separator **5**. In general, the electrolyte **6** is a substance that disassociates into electrically charged ions. A solvent that dissolves the substance may be included in some embodiments of the electrolyte **6**, as appropriate. The electrolyte **6** conducts electricity by ionic transport.

(80) In some embodiments, the electrolyte **6** may be in gelled or solid form (e.g., an ionic liquid impregnated polymer layer). Examples of such electrolytes are provided in International Publication No. WO 2015/102716 entitled “ADVANCED ELECTROLYTES FOR HIGH TEMPERATURE ENERGY STORAGE DEVICE” and published Jul. 9, 2015.

(81) In other embodiments, the electrolyte **6** may be in non-aqueous liquid form, e.g., an ionic liquid, e.g., of a type suitable for high temperature applications. Examples of such electrolytes are provided in International Publication No. WO 2015/102716 entitled “ADVANCED ELECTROLYTES FOR HIGH TEMPERATURE ENERGY STORAGE DEVICE” and published Jul. 9, 2015.

(82) In some embodiments, the storage cell **12** is formed into one of a wound form or prismatic

form which is then packaged into a cylindrical or prismatic housing 7. Once the electrolyte 6 has been included, the housing 7 may be hermetically sealed. In various examples, the package is hermetically sealed by techniques making use of laser, ultrasonic, and/or welding technologies. In addition to providing robust physical protection of the storage cell 12, the housing 7 is configured with external contacts to provide electrical communication with respective terminals 8 within the housing 7. Each of the terminals 8, in turn, provides electrical access to energy stored in the energy storage media 1, generally through electrical leads which are coupled to the energy storage media 1.

(83) As discussed herein, “hermetic” refers to a seal whose quality (i.e., leak rate) is defined in units of “atm-cc/second,” which means one cubic centimeter of gas (e.g., He) per second at ambient atmospheric pressure and temperature. This is equivalent to an expression in units of “standard He-cc/sec.” Further, it is recognized that 1 atm-cc/sec is equal to 1.01325 mbar-liter/sec. Generally, the ultracapacitor 10 disclosed herein is capable of providing a hermetic seal that has a leak rate no greater than about 5.0×10^{-6} atm-cc/sec, and may exhibit a leak rate no higher than about 5.0×10^{-10} atm-cc/sec. It is also considered that performance of a successfully hermetic seal is to be judged by the user, designer or manufacturer as appropriate, and that “hermetic” ultimately implies a standard that is to be defined by a user, designer, manufacturer or other interested party.

(84) Leak detection may be accomplished, for example, by use of a tracer gas. Using tracer gas such as helium for leak testing is advantageous as it is a dry, fast, accurate and non destructive method. In one example of this technique, the ultracapacitor 10 is placed into an environment of helium. The ultracapacitor 10 is subjected to pressurized helium. The ultracapacitor 10 is then placed into a vacuum chamber that is connected to a detector capable of monitoring helium presence (such as an atomic absorption unit). With knowledge of pressurization time, pressure and internal volume, the leak rate of the ultracapacitor 10 may be determined.

(85) In some embodiments, at least one lead (which may also be referred to herein as a “tab”) is electrically coupled to a respective one of the current collectors 2. A plurality of the leads (accordingly to a polarity of the ultracapacitor 10) may be grouped together and coupled to into a respective terminal 8. In turn, the terminal 8 may be coupled to an electrical access, referred to as a “contact” (e.g., one of the housing 7 and an external electrode (also referred to herein for convention as a “feed-through” or “pin”)). Suitable exemplary designs are provided in International Publication No. WO 2015/102716 entitled “ADVANCED ELECTROLYTES FOR HIGH TEMPERATURE ENERGY STORAGE DEVICE” and published Jul. 9, 2015.

(86) Various forms of the ultracapacitor 10 may be joined together. The various forms may be joined using known techniques, such as welding contacts together, by use of at least one mechanical connector, by placing contacts in electrical contact with each other and the like. A plurality of the ultracapacitors 10 may be electrically connected in at least one of a parallel and a series fashion.

(87) As used herein the symbol “wt %” means weight percent. For example, when referring to the weight percent of a solute in a solvent, “wt %” refers to the percentage of the overall mass of the solute and solvent mixture made up by the solute.

(88) The entire contents of each of the publications and patent applications mentioned above are incorporate herein by reference. In the event that the any of the cited documents conflicts with the present disclosure, the present disclosure shall control.

(89) While the invention has been described with reference to exemplary embodiments, it will be understood that various changes may be made and equivalents may be substituted for elements thereof without departing from the scope of the invention. For example, in some embodiments, one of the foregoing layers may include a plurality of layers there within. In addition, many modifications will be appreciated to adapt a particular instrument, situation or material to the teachings of the invention without departing from the essential scope thereof. Therefore, it is

intended that the invention not be limited to the particular embodiment disclosed as the best mode contemplated for carrying out this invention, but that the invention will include all embodiments falling within the scope of the appended claims.

Claims

1. An electrode for an energy storage device, the electrode comprising: a conductive layer; an active layer of energy storage media substantially free from binding agents and comprising a matrix of carbon nanotubes defining void spaces and binding carbonaceous material in the void spaces; and an adhesion layer being substantially free of binder material and disposed between the active layer and the conductive layer; wherein the matrix of carbon nanotubes comprises a concentration of carbon nanotubes that exceeds a percolation threshold thus causing the electrode to exhibit electrical connectivity within the active layer; wherein the matrix of carbon nanotubes is less than 1% by weight of the active layer; and wherein the carbon nanotubes comprise aligned carbon nanotubes.
 2. The electrode as in claim 1, wherein the matrix of carbon nanotubes comprises conductive paths configured to facilitate current flow within and through the active layer.
 3. The electrode as in claim 1, wherein the carbon nanotubes (CNT) comprise at least one of single wall nanotubes (SWNT), double wall nanotubes (DWNT), multiwall nanotubes (MWNT) and vertically aligned nanotubes.
 4. The electrode as in claim 1, wherein the conductive layer comprises a conductive material, and wherein the conductive material comprises a metal foil.
 5. The electrode as in claim 1, wherein a surface of the conductive layer is at least one of patterned, roughened and textured.
 6. The electrode as in claim 1, wherein a surface of the conductive layer comprises a nanostructured surface that promotes adhesion thereto.
 7. An energy storage device comprising the electrode as in claim 1, wherein the energy storage device is one of an ultracapacitor and a battery.
 8. The electrode as in claim 1, wherein the active layer comprises at least ninety (90) weight percent carbonaceous material.
 9. The electrode as in claim 1, wherein at least one of the active layer and the adhesion layer is a compressed layer substantially free from impurities.
 10. An energy storage device comprising: an electrode comprising a conductive layer, an adhesion layer disposed on the conductive layer, and an active layer disposed on the adhesion layer; the conductive layer comprising a metal foil that comprises at least one of a roughened, patterned, textured and nanostructured surface for promoting adhesion of the adhesion layer; the adhesion layer being substantially free of binder material and comprising carbonaceous material and an adhesion layer matrix of at least fifty percent aligned carbon nanotubes by weight of the adhesion layer, the carbonaceous material being bound in void spaces defined by the adhesion layer matrix; and the active layer being substantially free of binder material and comprising carbonaceous material and an active layer matrix of less than one percent aligned carbon nanotubes by weight of the active layer, the carbonaceous material being bound in void spaces defined by the active layer matrix, the active layer matrix comprising a concentration of carbon nanotubes that exceeds a percolation threshold thus causing the electrode to exhibit electrical connectivity within the active layer; wherein the active layer is thicker than the adhesion layer.
 11. The energy storage device as in claim 10, which is one of a battery and an ultracapacitor.
 12. The energy storage device as in claim 10, wherein at least one of the active layer and the adhesion layer is a compressed layer substantially free from impurities.
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